

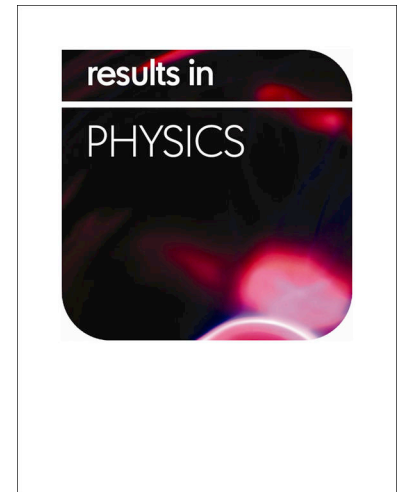
A novel method for the definition of central wavelength and spectral bandwidth

Hong Wang, Jianfeng Yang, Bin Xue

PII: S2211-3797(20)31794-0
DOI: <https://doi.org/10.1016/j.rinp.2020.103327>
Reference: RINP 103327

To appear in: *Results in Physics*

Received Date: 1 June 2020
Revised Date: 31 July 2020
Accepted Date: 11 August 2020



Please cite this article as: Wang, H., Yang, J., Xue, B., A novel method for the definition of central wavelength and spectral bandwidth, *Results in Physics* (2020), doi: <https://doi.org/10.1016/j.rinp.2020.103327>

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

A novel method for the definition of central wavelength and spectral bandwidth

Hong Wang^{a,b*}, Jianfeng Yang^a, and Bin Xue^{a*}

^a *Xi'an Institute of Optics and Precision Mechanics of Chinese Academy of Sciences, Xi'an, Shaanxi, China;*

^b *University of Chinese Academy of Sciences, Beijing, China.*

*Corresponding author: hoongwang@163.com, xuebin@opt.ac.cn

Abstract: Spectral calibration is an essential step in the calibration process of multispectral camera for Mars exploration. The precision of spectral calibration has a direct impact on the accuracy of Mars multispectral data inversion. The most common method for determining the central wavelength and spectral bandwidth of a multispectral camera is to fit the spectral response using a Gaussian function. The peak value fitted by Gauss is defined as the central wavelength of the camera, and the full width of the spectral band at half maximum is defined as the spectral bandwidth. But the spectral response of the Mars exploration multispectral camera in the near-infrared waveband deviates from the Gaussian distribution. In this case, the relative output deviation and the relative radiation error are large when data inversion is performed using the central wavelength and the spectral bandwidth determined by Gaussian fitting. In this paper, the spectral response of a multispectral camera is taken as a weight function, and a generalized weighted method is proposed to redefine the central wavelength and spectral bandwidth of the multispectral camera. The central wavelength is defined as the weighted average of wavelengths against the spectral response weighted factor, while the spectral bandwidth is defined as the generalized w bandwidth when the area within the bandwidth is w% of the total area. The simulation results showed that the relative output deviation was reduced by an average of 0.1% and a maximum of 1.3%, while the relative radiation error by an average of 0.14% and by a maximum of 1.83%, when compared with the Gaussian definition. The experiment results of the Mars exploration multispectral camera showed that there was an average reduction of 0.06% in relative output deviation and 0.05% in relative radiation error. The generalized weighted method provides another way for the definition of central wavelength and spectral bandwidth, and improves the precision of Mars multispectral data inversion.

Key words: multispectral; spectral calibration; spectral response; Gaussian; weight function

1. Introduction

At present, Mars exploration like a raging fire is carrying on, and China successfully reflected the first Mars probe on July 23 ^[1,2]. The multispectral camera is an important payload for Mars exploration ^[3], and provides visible and near-infrared images of the Martian surface, from which crucial observations of landscape morphology, geology, and atmospheric science may be derived ^[4]. The filter wheel of the Mars exploration multispectral camera in China has 8 geology filter locations ^[5], marked as B1~B8,

whose central wavelength and spectral bandwidth are 480(20), 525(20), 650(12), 700(15), 800(25), 900(30), 950(50) and 1000(50). In front of the brackets is the central wavelength, followed by the spectral bandwidth, all in nanometers.

The solar radiation reflected by targets on the Martian surface or self-radiation enters the multispectral camera and filters through various spectral channels to output multispectral image data [6,7]. The spectral features of the target can be extracted from these multispectral image data, and the information of the target composition and the existing state can be explained [8]. It is obvious that the development and application of the multispectral camera is aimed at accurately retrieving the geometrical and physical characteristics of objects on the surface of Mars through two-dimensional high-resolution images and one-dimensional spectral information, finally realizing the quantitative analysis and distinguishing the features of the objects [9,10]. In order to extract spectral information from the data obtained by the multispectral camera [11], it is necessary to first determine the meaning of the data collected by the multispectral camera [12], which mainly includes the central wavelength and spectral bandwidth of the camera. Therefore, after the fabrication of the multispectral camera, spectral calibration must be carried out to calibrate the central wavelength and spectral bandwidth of each spectral band, otherwise its use will be meaningless [13].

The most commonly used calibration method for multispectral camera is wavelength scanning method [14,15]. The spectral response of the camera can be obtained by the characteristic that monochromator can be scanned in its spectral range with a certain step, which is the function of the wavelength [16]. When determining the central wavelength and spectral bandwidth from the spectral response, the most common method is to fit the spectral response using a Gaussian function [17,18]. The peak position of the Gaussian function is defined as the central wavelength of the camera, the full width of the spectral band at the half peak is defined as the spectral bandwidth. However, some spectral responses are not close to Gaussian distribution. For example, the spectral response of the Chinese Mars multispectral camera in the near-infrared band rapidly diminishes after reaching its maximum, producing a trailing phenomenon. At this point, the Gaussian fit curve deviates from the actual spectral response. The target radiation at the central wavelength defined by this method has a large deviation from the camera output.

In this paper, the relationship between the camera output and the spectral response is analyzed. The spectral response is taken as a weight function. The weighted average of the wavelength relative

to the spectral response weight factor is taken as the central wavelength, which is called the weighted central wavelength. For the spectral bandwidth of multispectral camera, a scaling factor w is introduced to define the left and right wavelength positions of the bandwidth. The integral area of the spectral response at the left or right bandwidth is $\frac{w}{2}\%$ of the total area. The central wavelength and spectral bandwidth of the multispectral camera obtained in this paper will provide data support for the inversion of the Martian surface mineral reflectance spectra.

2. Methods

The theoretical output of the multispectral camera at the l -band can be regarded as the multispectral sampling value of the target spectral radiance $S(\lambda)$ to the spectral response $R_l(\lambda)$, which is denoted by b_l .

$$b_l = \int_0^{\infty} S(\lambda) \times R_l(\lambda) d\lambda \quad (1)$$

If the spectral response of each spectral band is a pulse function at the central wavelength Λ_l , then Eq. (1) can be expressed as,

$$b_l = \int_0^{\infty} \delta(\lambda - \Lambda_l) \times S(\lambda) d\lambda = S(\Lambda_l) \quad (2)$$

At this point, the camera output value b_l corresponds to the radiance $S(\Lambda_l)$ of the target at the central wavelength without deviation, which is ideal. The spectral radiance $S(\lambda)$ can be reconstructed from L data points $(\Lambda_1, b_1), \dots, (\Lambda_L, b_L)$. But in fact, the spectral response function of the multispectral camera at the l -band has a certain bandwidth Δ_l . Suppose that the spectrum radiance $S(\lambda)$ of the target can be approximated by the second order Taylor expansion at Λ_l , and the Eq. (1) should be expressed as,

$$b_l = S(\Lambda_l) + S'(\Lambda_l) \int_0^{\infty} R_l(\lambda) (\lambda - \Lambda_l) d\lambda + \frac{1}{2} S''(\Lambda_l) \int_0^{\infty} R_l(\lambda) (\lambda - \Lambda_l)^2 d\lambda \quad (3)$$

Where $S'(\Lambda_l)$ and $S''(\Lambda_l)$ are the first and second derivatives of the radiation spectrum $S(\lambda)$ at the central wavelength Λ_l , respectively. Let the second term on the right of the Eq. (3) be equal to 0, with the following equation,

$$\Lambda_l = \frac{\int_0^{\infty} R_l(\lambda) \times \lambda \times d\lambda}{\int_0^{\infty} R_l(\lambda) \times d\lambda} \quad (4)$$

The central wavelength in Eq. (4) is the weighted average of the wavelengths for the spectral response. The spectral response acts as a weight function, and its magnitude at a given wavelength determines the contribution of that wavelength to the central wavelength. The central wavelength is called the weighted central wavelength.

The deviation between the output of the multispectral camera and the radiance of the target at center wavelength can be expressed in terms of $b - S(\Lambda)$, which is related to the spectral response, bandwidth of the channel and the second derivative of spectral radiance at center wavelength.

$$b_l - S(\Lambda_l) = \frac{1}{2} S''(\Lambda_l) \times \int_0^\infty R_l(\lambda) \times (\lambda - \Lambda_l)^2 \times d\lambda \quad (5)$$

The wavelength positions on the left and right sides of spectral bandwidth at the l -band are recorded as λ_l^{left} and λ_l^{right} , respectively. The integral area of the spectral response in the left or right bandwidth as a percentage of the total area is $\frac{w}{2}\%$, and there is the equation as follows,

$$\begin{cases} \int_{\lambda_l^{left}}^{\Lambda_l} R_l(\lambda) d\lambda \bigg/ \int_0^\infty R_l(\lambda) d\lambda = \frac{w}{2}\% \\ \int_{\Lambda_l}^{\lambda_l^{right}} R_l(\lambda) d\lambda \bigg/ \int_0^\infty R_l(\lambda) d\lambda = \frac{w}{2}\% \end{cases} \quad (6)$$

Spectral bandwidth is defined as Eq. (7), which is called the generalized w bandwidth.

$$\Delta_l = \lambda_l^{right} - \lambda_l^{left} \quad (7)$$

The above definition method of central wavelength and spectral bandwidth is called generalized weighted method (GW).

In the application of the spectral response function of multispectral camera, the spectral response function of each band is sometimes not provided, only the central wavelength and spectral bandwidth are provided. In this case, the spectral response function of a multispectral camera is generally simulated using the Gaussian function [19]. Therefore, in order to compare the accuracy of the central wavelength and spectral bandwidth determined by the two methods, an ideal Gaussian spectral response is constructed according to the central wavelength and bandwidth calculated by the two definitions. Then the relative output deviation between the camera outputs at the ideal Gauss spectral responses and the actual spectral response is calculated. The smaller the relative output deviation is, the closer the constructed Gaussian spectral response is to the actual response.

$$\begin{cases} \Delta b_l^{gauss} = |b_l - b_l^{gauss}| / b_l \times 100\% \\ \Delta b_l^{gw} = |b_l - b_l^{gw}| / b_l \times 100\% \end{cases} \quad (8)$$

Where, Δb_l is the relative output deviation, and b_l^{gauss} and b_l^{gw} are the camera outputs of the proposed method and the Gaussian definition.

In addition, the aim of the multispectral camera is to retrieve the target radiation spectrum from the acquired multispectral image data. The offset between the output value of the multispectral camera and the radiance of the target at the central wavelength will affect the accuracy of the inversion [20,21]. The deviation caused by the generalized weighted method can be expressed in $b_l - S(\lambda_l)$, but the Gauss's cannot be expressed in a formula. For this purpose, the output value of the multispectral camera and the target's radiance at the Gaussian central wavelength and the weighted central wavelength (denoted by $S(\lambda_l^{gauss})$ and $S(\lambda_l^{gw})$ respectively) are simulated by using MATLAB. the relative radiation error ΔS between the output and radiance is calculated. The smaller the relative radiation error is, the more accurate the definition of central wavelength is.

$$\begin{cases} \Delta S_l^{gauss} = |b_l - S(\lambda_l^{gauss})| / b_l \times 100\% \\ \Delta S_l^{gw} = |b_l - S(\lambda_l^{gw})| / b_l \times 100\% \end{cases} \quad (9)$$

3. Results and Analysis

3.1 Simulation

To compare the accuracy of the generalized weighted method and the Gaussian method in defining the central wavelength and spectral bandwidth, and their effects on spectral reconstruction, eight spectral response in the spectral range of 610nm ~ 690nm were simulated by MATLAB, as shown in Fig. 1. Among them, ① was an ideal Gaussian distribution, ②~⑧ was a non-Gaussian distribution, but ②~④ was symmetric. In order to cover the entire spectral range of the Mars exploration multispectral camera, these spectral responses were replicated within the seven other spectral ranges (440nm~520nm, 485nm~565nm, 610nm~690nm, 660nm~740nm, 760nm~840nm, 860nm~940nm, 910~990nm, 960~1040nm). There were eight bands including the former bands. The central wavelengths of these eight bands corresponded to the central wavelengths of the eight spectral channels of the Mars exploration multispectral camera.

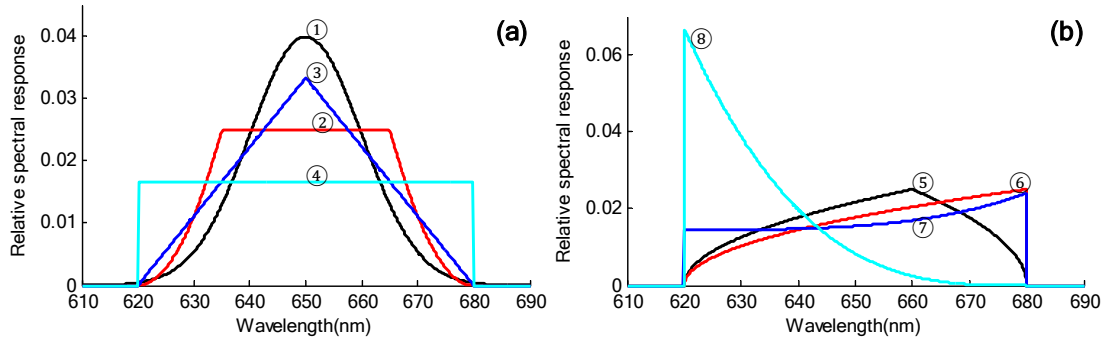


Fig. 1 Analog spectral response

The area under the spectral bandwidth defined by Gaussian method accounted for 76.2% of the total area. In order to compare the accuracy of the two methods under the same area ratio, the value of w in the generalized weighted method was also set to 76.2%.

The central wavelengths and spectral bandwidths calculated by the two methods were shown in Table 1. It can be seen that when the spectral response was Gaussian distribution, the central wavelength and the spectral bandwidth calculated by the two definitions were the same; when the spectral response was non-Gaussian but symmetric, the central wavelength calculated by the two definitions was the same but the spectral bandwidth was not the same; when the spectral response was asymmetrically distributed, the results of the two calculations differed greatly.

Table 1 Central wavelength and spectral bandwidth

Spectral response	Central wavelength (nm)		Spectral bandwidth (nm)	
	Gauss	GW	Gauss	GW
1	650.0	650.0	23.5	23.4
2	650.0	650.0	34.0	30.4
3	650.0	650.0	31.2	30.6
4	650.0	650.0	53.7	45.4
5	653.2	652.0	39.8	37.5
6	654.4	652.4	55.8	46.8
7	658.7	656.0	44.7	41.9
8	628.0	632.0	17.7	31.5

In the spectral range of 610nm ~ 690nm, the Gaussian spectral responses constructed from the central wavelength and spectral bandwidth defined by the two methods were shown in Fig. 2.

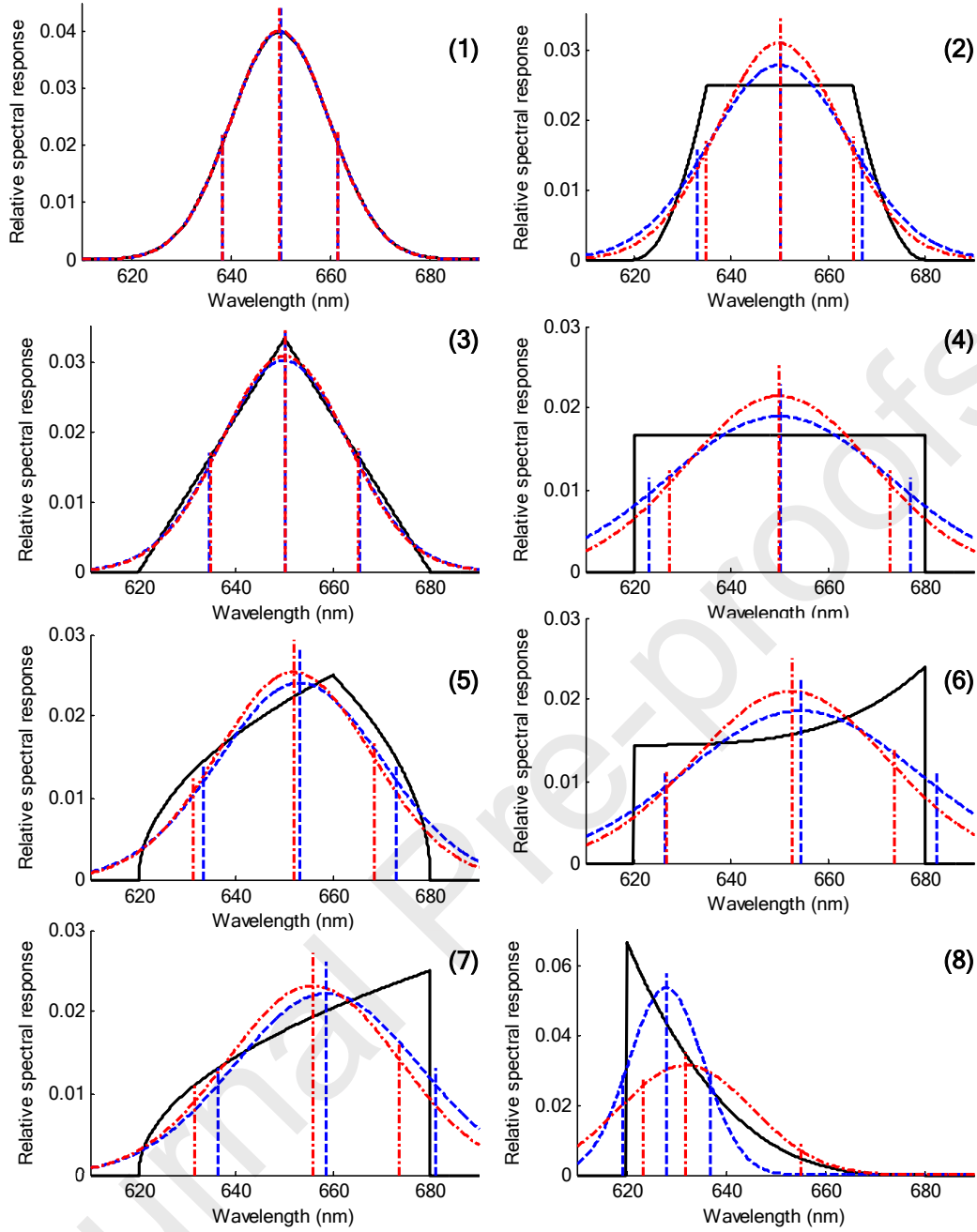


Fig. 2 Actual spectral response and constructed Gaussian response

Note: From top to bottom, left to right, corresponding to spectral response ①~⑧; —, the actual spectral response; - - - - -, the constructed Gaussian response based on the Gaussian method; - · - · -, the constructed Gaussian response based on the generalized weighted method.

Next, spectral radiance data of 19 Martian surface minerals were selected to calculate the relative output deviation and relative radiation error. The Radiation Spectra of the 19 minerals were shown in Fig. 3. The average relative output deviation for the 19 minerals was shown in Table 2.

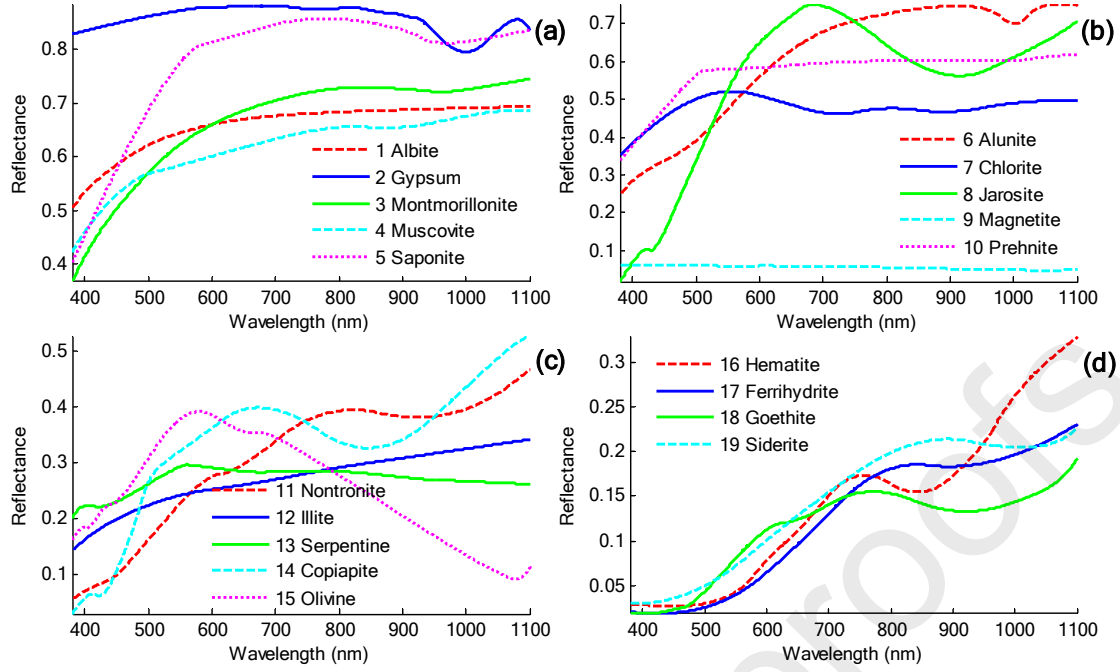


Fig. 3 Radiation Spectra of 19 Martian surface minerals

Table 2 Average relative output deviation

Spectral range (nm)	Relative output deviation	Spectral response							
		1	2	3	4	5	6	7	8
440~520	Δb^{gauss}	0	0.14%	0.06%	0.16%	0.42%	0.27%	0.48%	2.21%
	Δb^{gw}	0	0.08%	0.04%	0.08%	0.11%	0.24%	0.38%	0.91%
485~565	Δb^{gauss}	0	0.19%	0.08%	0.18%	0.41%	0.27%	0.46%	1.60%
	Δb^{gw}	0	0.06%	0.06%	0.05%	0.08%	0.25%	0.37%	0.67%
610~690	Δb^{gauss}	0	0.06%	0.02%	0.05%	0.15%	0.09%	0.17%	0.67%
	Δb^{gw}	0	0.02%	0.02%	0.01%	0.03%	0.09%	0.14%	0.23%
660~740	Δb^{gauss}	0	0.04%	0.02%	0.04%	0.15%	0.09%	0.17%	0.70%
	Δb^{gw}	0	0.02%	0.01%	0.01%	0.04%	0.10%	0.16%	0.30%
760~840	Δb^{gauss}	0	0.05%	0.02%	0.04%	0.09%	0.05%	0.10%	0.48%
	Δb^{gw}	0	0.02%	0.01%	0.01%	0.04%	0.06%	0.11%	0.19%
860~940	Δb^{gauss}	0	0.06%	0.02%	0.05%	0.10%	0.07%	0.10%	0.28%
	Δb^{gw}	0	0.04%	0.02%	0.02%	0.03%	0.04%	0.06%	0.12%
910~990	Δb^{gauss}	0	0.04%	0.02%	0.03%	0.12%	0.07%	0.14%	0.46%
	Δb^{gw}	0	0.02%	0.01%	0.03%	0.03%	0.09%	0.13%	0.21%
960~1040	Δb^{gauss}	0	0.08%	0.03%	0.05%	0.11%	0.05%	0.11%	0.55%
	Δb^{gw}	0	0.02%	0.02%	0.04%	0.03%	0.10%	0.13%	0.19%

It can be seen from Table 2 that when the spectral response was Gaussian distribution, the relative output deviations of the two definitions were the same; when the spectral response was not Gaussian distribution, the relative output deviation of the generalized weighted method was less than that of the Gaussian method. The average relative output deviation of the generalized weighted definition was

0.20%, and that of the Gaussian definition was 0.10%, with an average reduction of 0.10% and a maximum reduction of 1.30%. This reduction was related to the spectrum radiance distribution of the target and the extent to which the spectral response deviated from the Gaussian distribution. The closer the constructed Gaussian spectral response was to the actual spectral response, the lower the introduced relative output deviation was. This goes some way to explaining that the Gaussian spectral response constructed from data obtained by the generalized weighted method was closer to the actual spectral response than the Gaussian fitting.

The relative radiation errors corresponding to the two definitions were calculated below. MATLAB was used to simulate the output value of the camera at the lth spectral band of the mth mineral in Fig. 3, which was denoted by $b_{m,l}$, and the radiance at the Gaussian central wavelength and the weighted central wavelength, which were denoted by $S_m(\lambda_l^{gauss})$ and $S_m(\lambda_l^{gw})$, respectively. The average relative radiation error ΔS between the camera value and the radiance was calculated. The results were shown in Table 3.

Table 3 Average relative radiation error

Spectral range (nm)	Relative radiance error	Spectral response							
		1	2	3	4	5	6	7	8
440~520	ΔS^{gauss}	0.49%	0.64%	0.63%	1.07%	1.36%	1.70%	1.71%	2.77%
	ΔS^{gw}	0.49%	0.64%	0.63%	1.07%	0.93%	1.30%	1.18%	0.94%
485~565	ΔS^{gauss}	0.46%	0.63%	0.64%	1.17%	0.62%	1.14%	0.90%	1.96%
	ΔS^{gw}	0.46%	0.63%	0.64%	1.17%	0.64%	0.95%	0.96%	0.69%
610~690	ΔS^{gauss}	0.15%	0.20%	0.20%	0.37%	0.35%	0.49%	0.51%	0.71%
	ΔS^{gw}	0.15%	0.20%	0.20%	0.37%	0.26%	0.37%	0.26%	0.21%
660~740	ΔS^{gauss}	0.15%	0.19%	0.19%	0.29%	0.32%	0.48%	0.56%	0.88%
	ΔS^{gw}	0.15%	0.19%	0.19%	0.29%	0.21%	0.30%	0.24%	0.18%
760~840	ΔS^{gauss}	0.10%	0.13%	0.13%	0.26%	0.29%	0.46%	0.53%	0.58%
	ΔS^{gw}	0.10%	0.13%	0.13%	0.26%	0.18%	0.27%	0.23%	0.14%
860~940	ΔS^{gauss}	0.12%	0.16%	0.15%	0.20%	0.16%	0.22%	0.25%	0.35%
	ΔS^{gw}	0.12%	0.16%	0.15%	0.20%	0.15%	0.21%	0.22%	0.26%
910~990	ΔS^{gauss}	0.28%	0.35%	0.34%	0.49%	0.33%	0.41%	0.42%	0.53%
	ΔS^{gw}	0.28%	0.35%	0.34%	0.49%	0.39%	0.48%	0.31%	0.20%
960~1040	ΔS^{gauss}	0.27%	0.36%	0.35%	0.58%	0.50%	0.71%	0.63%	0.59%
	ΔS^{gw}	0.27%	0.36%	0.35%	0.58%	0.45%	0.60%	0.48%	0.15%

From Table 3, it can be seen that when the spectral response was a Gaussian distribution or a symmetric distribution, the central wavelengths of the two definitions were the same, resulting in the

same relative radiation error, which was mainly related to the spectral bandwidth. And if the spectral response was asymmetric, the larger the asymmetric distribution was, the larger the relative radiation error was. The relative error of the generalized weighted definition was much lower than that of the Gaussian definition. The average relative error of the generalized weighted definition was 0.55%, and that of the Gaussian definition was 0.41%, with an average reduction of 0.14% and a maximum reduction of 1.83%. This indicated that the multispectral camera output value deviated less from the radiance value of the target at the generalized central wavelength, and the target radiation spectrum obtained by the inversion of this central wavelength was more accurate.

3.2 Experiment

The spectral responses of the eight spectral channels of the Mars exploration multispectral camera obtained by using wavelength scanning method were shown in Fig. 4.

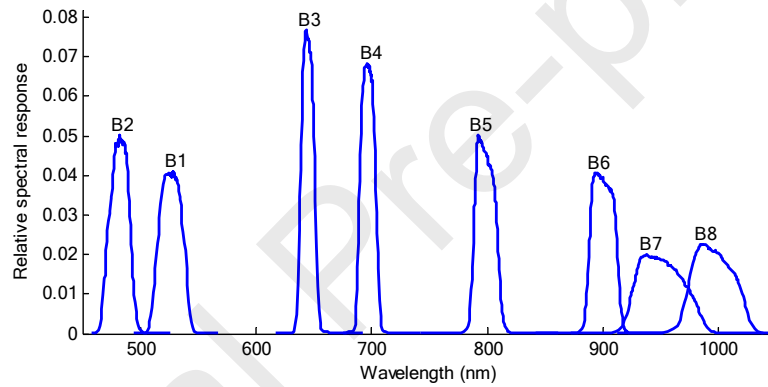


Fig. 4 Spectral response of Mars exploration multispectral camera

The central wavelength and spectral bandwidth defined by the two methods were shown in Table 4. It can be seen that the spectral response of B1~B4 was close to the Gaussian distribution, and the central wavelength and spectral bandwidth calculated by the two methods were close to each other. In the channel of B5 ~ B8, the spectral response appeared a dragging tail at the long wavelength and deviated from the Gaussian distribution. There were some differences between the two methods in calculation of the central wavelength and spectral bandwidth. The maximum deviation of central wavelength was about 1 nm, and the maximum deviation of spectral bandwidth was 2.5 nm.

Table 4 Central wavelength and spectral bandwidth of Mars exploration multispectral camera

Band	Central wavelength (nm)		Spectral bandwidth (nm)	
	GFM	GWM	GFM	GWE
B1	526.4	526.4	21.8	20.1
B2	482.1	481.9	18.3	17.2
B3	644.5	644.8	11.5	10.8
B4	696.6	696.8	12.9	11.9
B5	797.9	798.4	18.5	17.0
B6	899.6	899.9	21.9	19.7
B7	949.1	950.5	46.3	44.0
B8	994.6	995.5	41.1	38.6

The relative output deviations and relative radiation errors based on the two methods were shown in Tables 5 and 6 respectively. As can be seen from Tables 5 and 6, the relative output deviation and the relative radiation error of the generalized weighted method were all smaller than those of the Gaussian method, with the average decreases of 0.06% and 0.05%, respectively. This demonstrated the reliability of the generalized weighted method in the practical application of spectral calibration of multispectral camera.

Table 5 Average relative output deviation of Mars exploration multispectral camera

Relative output deviation	Band							
	B1	B2	B3	B4	B5	B6	B7	B8
Δb^{gauss}	0.07%	0.11%	0.05%	0.04%	0.06%	0.03%	0.15%	0.14%
Δb^{gw}	0.02%	0.02%	0.01%	0.01%	0.01%	0.01%	0.03%	0.02%

Table 6 Average relative radiation error of Mars exploration multispectral camera

Relative radiance error	Band							
	B1	B2	B3	B4	B5	B6	B7	B8
ΔS^{gauss}	0.26%	0.56%	0.12%	0.10%	0.13%	0.11%	0.58%	0.41%
ΔS^{gw}	0.26%	0.46%	0.12%	0.08%	0.10%	0.09%	0.44%	0.32%

4. Conclusions

For the spectral calibration of the Mars exploration multispectral camera, a new definition method of central wavelength and spectral bandwidth is presented. The spectral response is taken as a weight function, the central wavelength is defined as the weighted average of wavelengths against the spectral response weighted factor, and the spectral bandwidth is defined as the generalized w bandwidth when the area within the bandwidth is w% of the total area. The simulation and experimental results showed

that the relative output deviation of the proposed method was lower than that of the Gaussian method, and the relative radiation error was also reduced. This proposed method is not only suitable for acquiring the central wavelength and spectral bandwidth of the Mars exploration multispectral camera, but also suitable for some spectral responses that deviate heavily from the Gaussian distribution.

Acknowledgment

We wish to thank all the participants for their great works in Mars exploration Project, whose joint efforts have made the data acquisition and processing used for this study possible. And it is gratefully acknowledged that this work was supported by the National Natural Science Foundation of China (11941002).

Disclosure statement

No potential conflict of interest was reported by the author(s).

References

- [1] JIANLAN S. China Emphasizes International Cooperation in Future Lunar and Deep Space Exploration [J]. Bulletin of the Chinese Academy of Sciences, 2019, 2)
- [2] Yan G , Jishi Z , Sha L I , et al. A Brief Introduction of the First Mars Exploration Mission in China[J]. Journal of Deep Space Exploration, 2018.
- [3] VAGO J, WESTALL F, TEAMS L, et al. Habitability on Early Mars and the Search for Biosignatures with the ExoMars Rover [J]. Astrobiology, 2017, 17(471-510).
- [4] COATES A, JAUMANN R, GRIFFITHS A, et al. The PanCam Instrument for the ExoMars Rover [J]. Astrobiology, 2017, 17.
- [5] YINGZHUO J, YU F, YONGLIAO Z. Scientific Objectives and Payloads of Chinese First Mars Exploration [J]. Journal of Space Science, 2018, 5): 650-5.
- [6] BELL J, JOSEPH J K, SOHL-DICKSTEIN J, et al. In-flight calibration and performance of the Mars Exploration Rover Panoramic Camera (Pancam) instruments [J]. Journal of Geophysical Research, 2006, 111.
- [7] HUCK F O, BURCHER E E, TAYLOR E J, et al. Radiometric performance of the Viking Mars lander cameras [J]. 1975, 75.
- [8] BELL J F I, CALVIN W M, FARRAND W H, et al. The Martian Surface: Mars Exploration Rover Pancam multispectral imaging of rocks, soils, and dust at Gusev crater and Meridiani Planum [J]. 2008.

- [9] JF B, SQUYRES S W, ARVIDSON R E, et al. Pancam multispectral imaging results from the Spirit Rover at Gusev crater [J]. 2004, 305(5685): p.800-6.
- [10] III J F B, SQUYRES S W, ARVIDSON R E, et al. Pancam Multispectral Imaging Results from the Opportunity Rover at Meridiani Planum [J]. 2004, 306(5702): p.1703-9.
- [11] IV S, SODERBLOM J, FARRAND W, et al. Mars Exploration Rover Panoramic Camera Multidimensional Analyses and Surface Spectral Variability [J]. 2004, 35(2166).
- [12] BELL J F, SQUYRES S W, HERKENHOFF K E, et al. The NASA 2003 Mars Exploration Rover Panoramic Camera (Pancam) Investigation [M]. AGU Fall Meeting Abstracts. 2002: P22A-0395.
- [13] ZHOU F. Inversion and Accuracy Verification of Spectral Calibration Parameters for Spaceborne Infrared Hyperspectral Instrument [D]; Chinese Academy of Meteorological Sciences, 2019.
- [14] Sun C. The Study on Full-Field and Automatic Methodology of Spectral Calibration for Imaging Spectrometers [D]; Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, 2013.
- [15] SHU R, XUE Y-Q, YANG Y-D. Calibration and application of airborne pushbroom hyperspectral imager (PHI) [J]. Proceedings of SPIE - The International Society for Optical Engineering, 2004.
- [16] ZHANG K, YU K, LIU Y, et al. An improved algorithm for spectral emissivity measurements at low temperatures based on the multi-temperature calibration method [J]. International Journal of Heat and Mass Transfer, 2017, 114(1037-44).
- [17] Zhu M, Zhang M, Hu L-Z, et al. Peak search algorithm for center wavelength detection in fiber Bragg gratings [J]. Study on Optical Communications, 2011, 37(5): 60-3.
- [18] Guo C. Technology of Spectrogram Peak Detection and Implementation [D]; XiDian University, 2015.
- [19] BEIRLE S, LAMPEL J, LEROT C, et al. Parameterizing the instrumental spectral response function and its changes by a super-Gaussian and its derivatives [J]. Atmospheric Measurement Techniques, 2017, 10(581-98).
- [20] SQUYRES S, KNOLL A, ARVIDSON R, et al. Exploration of Victoria Crater by the Mars Rover Opportunity [J]. Science (New York, NY), 2009, 324(1058-61).
- [21] JEFFREY, R., JOHNSON, et al. Spectrophotometric properties of materials observed by Pancam on the Mars Exploration Rovers: 3. Sols 500–1525 [J]. Icarus, 2015.

Hong Wang: Conceptualization, Methodology, Data curation, Writing.

Jianfeng Yang: Supervision.

Bin Xue: Supervision, Reviewing, Validation.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Highlights

1. A new idea and method is used to analyze the relationship between spectral response and central wavelength and spectral bandwidth.
2. A new generalized weighted method is proposed, in which the spectral response is taken as a weight function to redefine the central wavelength and spectral bandwidth.
3. Based on generalized weighted method, the relative output deviation and the relative radiation error in data reconstruction are calculated.